



**CARINGBAH
HIGH
SCHOOL**

Name: _____

2020

**Preliminary
Task 3**

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 90 minutes
- Write using black pen
- NESA approved calculators and Mathomat may be used
- A reference sheet is provided
- In Questions 11–15, show relevant mathematical reasoning and/or calculations

**Total marks:
65**

Section I – 10 marks (pages 2–4)

- Attempt Questions 1–10
- Allow about 10 minutes for this section

Section II – 55 marks (pages 5–9)

- Attempt Questions 11–15
- Allow about 80 minutes for this section

Marker's Use Only

Question	1 – 10	11	12	13	14	15	Total	
Mark	/10	/11	/11	/11	/11	/11	/65	%

This paper must not be removed from the examination room

Section I**10 marks****Attempt Questions 1–10****Allow about 10 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10.

1 Which of the following is a factor of $P(x) = x^3 + x^2 + x + 1$?

(A) $(x - 1)$

(B) $(x + 1)$

(C) $\left(x - \frac{1}{2}\right)$

(D) $\left(x + \frac{1}{2}\right)$

2 What are the integers x and y such that $\frac{\sqrt{3}}{2\sqrt{3}+3} = x + y\sqrt{3}$?

(A) $x = -1$ and $y = 2$

(B) $x = 1$ and $y = -2$

(C) $x = 2$ and $y = -1$

(D) $x = -2$ and $y = 1$

3 A six-sided die has one side with a value of 1, two sides each with a value of 2 and three sides each with a value of 3. On rolling this die what would be the expected value?

(A) 2

(B) $2\frac{1}{3}$

(C) $2\frac{1}{2}$

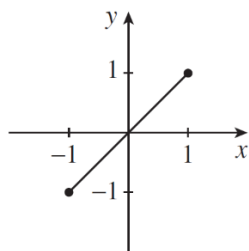
(D) $2\frac{2}{3}$

4 If $f(x) = x^2$ and $g(x) = \sqrt{x}$. Find $g(f(2))$.

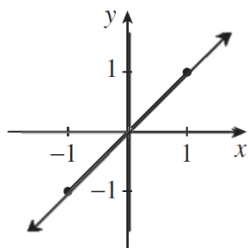
- (A) x
- (B) 0
- (C) 1
- (D) 2

5 Which of the following is the graph of $y = \sin^{-1}(\sin x)$?

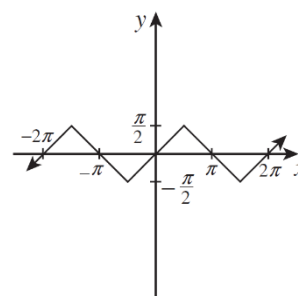
(A)



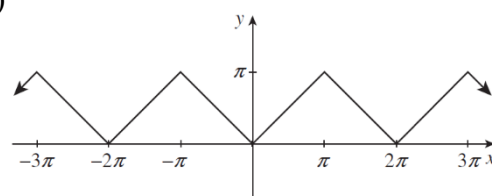
(B)



(C)



(D)



6 Which expression is equivalent to $\frac{2 \cos^2 \theta}{1 - \sin \theta}$?

- (A) $2 - 2 \sin \theta$
- (B) $2 + 2 \sin \theta$
- (C) $2 - 2 \cos \theta$
- (D) $2 - 2 \cos \theta$

- 7 For which value of a is this function continuous for all values of x ?

$$f(x) = \begin{cases} a - x^2, & x < 1 \\ (x - a)^2, & x \geq 1 \end{cases}$$

- (A) $a = 5$
(B) $a = 4$
(C) $a = 3$
(D) $a = 2$
- 8 What is the domain of the function $f(x) = \ln(3x - x^2)$?

- (A) $0 < x < 3$
(B) $0 \leq x < 3$
(C) $0 < x \leq 3$
(D) $0 \leq x \leq 3$

- 9 Which parametric equations represent a circle?

- (A) $x = 2t$ and $y = t^2$
(B) $x = 2t$ and $y = t + 2$
(C) $x = 2\cos(t)$ and $y = \sin(t)$
(D) $x = 2\cos(t) + 2\sin(t)$ and $y = 2\cos(t) - 2\sin(t)$

- 10 Three girls and three boys sit around a round table.

What is the probability that they sit boy-girl-boy-girl-boy-girl?

- (A) $\frac{1}{10}$ (C) $\frac{1}{120}$
(B) $\frac{3}{20}$ (D) $\frac{1}{20}$

Section II**55 marks****Attempt Questions 11–15****Allow about 80 minutes for this section**

Answer each question on SEPARATE writing paper. Extra writing booklets are available.

In Questions 11 – 15, your responses should include relevant mathematical reasoning and/or calculations.

		Marks
Question 11 (11 marks) Use SEPARATE writing paper.		
(a)	Solve: $ 2x - 1 \geq 3$	2
(b)	A bag contains three red, four yellow and five blue balls. If three balls are drawn from the bag simultaneously, find the probability that: (i) all three balls are red. (ii) At least one of the balls are yellow?	1 2
(c)	If $\sin x = \frac{2}{3}$, where $90^\circ \leq x \leq 180^\circ$ and $\cos y = \frac{3}{4}$, where $0^\circ \leq y \leq 90^\circ$, what is the exact value of $\sin(x + y)$?	3
(d)	If $P(x) = x(x^2 - 1)^2$, (i) Find all the roots of $P(x)$. (ii) Show that $P(1) = P'(1) = 0$.	1 2

End of Question 11

Question 12 (11 marks) Use SEPARATE writing paper.

- (a) Sketch the graph of $y = (1 - x)^3(2 + x)^2(3 - x)$ showing all the intercepts. **3**
- (b) Let $f(x) = (x - 1)^2$, for the domain $x > 1$.
- (i) Classify $f(x)$ as one-to-one or many-to-one. **1**
- (ii) Find any values of x for which $f(x) = f^{-1}(x)$. **2**
- (c) Find the multiplicity of the root $x = -1$ of the equation $P(x) = 0$ where $P(x) = x^4 + 4x^3 + 7x^2 + 6x + 2$. **2**
- (d) A certain chemical treatment of blue-green algae in a river causes it to decrease based on the equation $A = A_0 \times e^{kt}$, where A is the kilograms of algae and t is time in months. **3**
- If 250 kg of blue-green algae reduces to 150 kg after 3 months, find how long it will take, to the nearest month, to reduce the bluegreen algae to 10 kg.

End of Question 12

Question 13 (11 marks) Use SEPARATE writing paper.

- (a) Sketch the graph of the function $f(x) = -2 \tan^{-1} x$ showing the intercepts on the axes and the equations of the asymptotes. **2**
- (b) A polynomial of degree 4 has a double zero at 2 with the other two zeros being k and $-k$. **3**
When $x = 1$ it takes the value 8 and when $x = 3$ it takes the value 80.
Find the possible values of k .
- (c) If $f(x) = x^2 - 1$ sketch the graph of $y = \frac{1}{f(x)}$ indicating all intercepts and asymptotes. **3**
- (d) A Maths test is to consist of ten questions.
- (i) In how many ways can it be arranged so that the shortest question is first and the longest question is last? **1**
- (ii) What is the probability that the shortest and longest questions are next to one another? **2**

End of Question 13

Question 14 (11 marks) Use SEPARATE writing paper.

- (a) The numbers 1 to 20 are either odd or even as well as being prime, composite or neither. **2**
If an odd number is chosen at random what is the probability that it is also prime?

- (b) i) Find a simplified expression for $\operatorname{cosec} \theta (\cos \theta - 1)$ in terms of t where $t = \tan \frac{\theta}{2}$. **2**

- ii) Hence solve $\operatorname{cosec} \theta (\cos \theta - 1) = -1$ for $0^\circ \leq \theta \leq 180^\circ$. **1**

- (c) If α, β and γ are the roots of the cubic equation $2x^3 + 4x - 3 = 0$. **3**

Find the value of $\alpha^3 + \beta^3 + \gamma^3$.

- (d) Solve the inequation: **3**

$$\frac{x}{x-1} \geq 2$$

End of Question 14

Question 15 (11 marks) Use SEPARATE writing paper.

- (a) Solve: 3

$$2 \times 4^x - 3 \times 2^x + 1 = 0.$$

- (b) Find the number of ways in which the letters of the word TRIANGLE can be arranged in a line:

- i) so that all the vowels are next to each other? 1

- ii) so that consonants are in the two end positions? 1

- (c) Consider the function $f(x) = (x - 2)^2$, $0 \leq x \leq 2$.

- i) Find the inverse function $f^{-1}(x)$ and state its domain. 2

- ii) Sketch the graph of the inverse function $f^{-1}(x)$ showing the endpoints. 2

- (d) If $\sec x + \tan x = k$, find an expression for $\sec x$ in terms of k . 2

End of Exam

BLANK PAGE

Name: _____

Mathematics Extension 1

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct ↖

- Start ➔ 1. A ☐ B ☐ C ☐ D ☐
Here
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

Y11 Extension 1 2020 AT3 Exam Answers

Section I

Q1) $P(-1) = -1^3 + -1^2 - 1 + 1 = 0$

$\therefore (x + 1)$ is a factor.

(B)

Q2) $\frac{\sqrt{3}}{2\sqrt{3}+3} \times \frac{2\sqrt{3}-3}{2\sqrt{3}-3} = \frac{6-3\sqrt{3}}{12-9} = \frac{3(2-\sqrt{3})}{3} = 2 - \sqrt{3}$

$\therefore x = 2$ and $y = -1$

(C)

Q3) $E(X) = \left(1 \times \frac{1}{6}\right) + \left(2 \times \frac{2}{6}\right) + \left(3 \times \frac{3}{6}\right) = \frac{14}{6} = 2\frac{1}{3}$

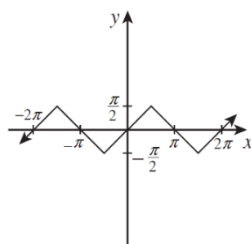
(B)

Q4) $g(f(x)) = \sqrt{(x^2)} = x \therefore g(f(2)) = 2$

(D)

Q5) $y = \sin^{-1}(\sin x)$

(C)



Q6) $\frac{2 \cos^2 \theta}{1 - \sin \theta} = \frac{2(1 - \sin^2 \theta)}{1 - \sin \theta} = \frac{2(1 - \sin \theta)(1 + \sin \theta)}{1 - \sin \theta} = 2 + 2 \sin \theta$

(B)

Q7) When $a = 2$, $f(x) = 2 - x^2$ for $x < 1$ and $(x - 2)^2$ for $x \geq 1$

which is "smooth" & continuous

(D)

Q8) Domain of $f(x) = \ln(3x - x^2)$ is $(0, 3)$

(A)

Q9) Parametric equations of a circle are

$x = 2 \cos(t) + 2 \sin(t)$ & $y = 2 \cos(t) - 2 \sin(t)$

(D)

Q10) $P(\text{Boy} - \text{girl} - \text{boy} - \text{girl}) = \frac{3!2!}{5!} = \frac{1}{10}$

(A)

Section II

Q11)

a) $|2x - 1| \geq 3 \therefore 2x - 1 \geq 3$ or $-2x + 1 \leq 3$

$\therefore x \geq 2$ or $x \leq -1$ or $(-\infty, -1] \cup [2, \infty)$

b)

i) $P(\text{all red}) = \frac{3}{12} \times \frac{2}{11} \times \frac{1}{10} = \frac{1}{220}$

ii) $P(\text{at least 1 yellow}) = 1 - \left(\frac{8}{12} \times \frac{7}{11} \times \frac{6}{10}\right) = \frac{41}{55}$

c) $\sin x = \frac{2}{3}$, where $90^\circ \leq x \leq 180^\circ \therefore \cos x = \frac{-\sqrt{5}}{3}$

$\cos y = \frac{3}{4}$, where $0^\circ \leq x \leq 90^\circ \therefore \sin y = \frac{\sqrt{7}}{4}$

$\therefore \sin(x + y) = \frac{2}{3} \times \frac{3}{4} - \frac{\sqrt{5}}{3} \times \frac{\sqrt{7}}{4} = \frac{6 - \sqrt{35}}{12} = \frac{1}{2} - \frac{\sqrt{35}}{12}$

d)

i) $P(x) = x(x^2 - 1)^2 \therefore \text{Roots } x = -1, 0, 1$

ii) $P(1) = 1(1^2 - 1)^2 = 0$

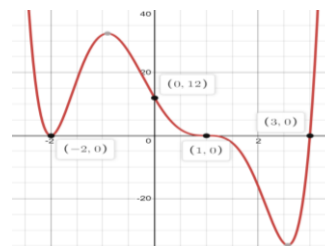
$P'(x) = (x^2 - 1)^2 \times 1 + x \times 2(x^2 - 1) \times 2x$
 $= (x^2 - 1)[(x^2 - 1) + 4x^2]$

$\therefore P'(1) = (1^2 - 1)[(1 - 1) + 4 \times 1^2] = 0$

$\therefore P(1) = P'(1) = 0$

Q12)

a) $y = (1 - x)^3(2 + x)^2(3 - x)$



b) $f(x) = 2(x - 1)^2$ for the domain $x > 1$

i) One-to-One

ii) $f(x) = f^{-1}(x)$ when $f(x) = x$

$\therefore 2(x - 1)^2 = x$

$\therefore 2x^2 - 4x + 2 = x$

$\therefore 2x^2 - 5x + 2 = 0$

$\therefore (2x - 1)(x - 2) = 0$

$\therefore x = 2$ only for the domain $x > 1$

c) $P(x) = x^4 + 4x^3 + 7x^2 + 6x + 2$

$\therefore P(-1) = -1^4 + 4 \cdot -1^3 + 7 \cdot -1^2 + 6 \cdot -1 + 2 = 0$

therefore $(x + 1)$ is a factor

$\therefore P'(x) = 4x^3 + 12x^2 + 14x + 6$

$\therefore P'(-1) = 4 \cdot -1^3 + 12 \cdot -1^2 + 14 \cdot -1 + 6 = 0$

therefore $(x + 1)^2$ is a factor

$\therefore P''(x) = 12x^2 + 24x + 14$

$\therefore P''(-1) = 12 \cdot -1^2 + 24 \cdot -1 + 14 \neq 0$

therefore multiplicity is 2

d) $A = 250 \times e^{kt}$

when $t = 3$, $A = 150 = 250 \times e^{k \times 3}$

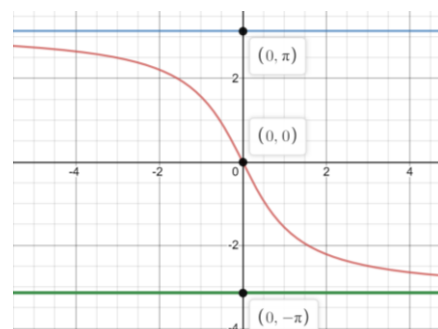
therefore $k = \ln\left(\frac{150}{250}\right) \div 3 \approx -0.17$

when $A = 10 = 250 \times e^{-0.17t}$,

$t = \ln\left(\frac{10}{250}\right) \div -0.17 \approx 19$ months

Q13)

a) $f(x) = -2 \tan^{-1} x$



Y11 Extension 1 2020 AT3 Exam Answers

$$= (-4(\alpha + \beta + \gamma) + 9)/2 = (-4(0) + 9)/2 = 9/2$$

Q13) (Continued)

b) $P(x) = a(x-2)^2(x-k)(x+k)$

$$P(1) = a(1-2)^2(1-k)(1+k) = 8$$

$$\therefore a(1-k^2) = 8 - \textcircled{1}$$

$$P(3) = a(3-2)^2(3-k)(3+k) = 80$$

$$\therefore a(9-k^2) = 80 - \textcircled{2}$$

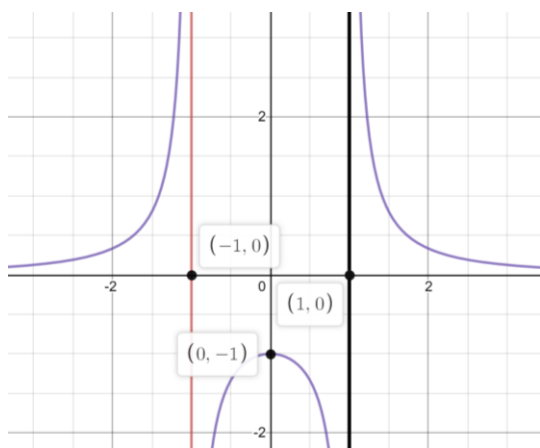
$$\therefore \textcircled{2} \div \textcircled{1} \frac{a(9-k^2)}{a(1-k^2)} = 10$$

$$\therefore (9-k^2) = 10(1-k^2)$$

$$\therefore 9-k^2 = 10-10k^2$$

$$\therefore 9k^2 = 1, k = \pm \frac{1}{3}$$

c) $f(x) = x^2 - 1 \therefore y = \frac{1}{f(x)}$



d)

i) $1 \times 1 \times 8! = 40,320$

ii) $P = \frac{9!2!}{10!} = \frac{1}{5}$

Q14)

a) odds = 10 primes which are odd = 7

$$\therefore P = \frac{7}{10}$$

b)

i) $\operatorname{cosec} \theta (\cos \theta - 1) = \frac{1+t^2}{2t} \times \left(\frac{1-t^2}{1+t^2} - 1 \right)$

$$= \frac{1+t^2}{2t} \times \left(\frac{1-t^2}{1+t^2} - \frac{1+t^2}{1+t^2} \right) = \frac{1+t^2}{2t} \times \left(\frac{1-t^2-1-t^2}{1+t^2} \right)$$

$$= \frac{1+t^2}{2t} \times \left(\frac{-2t^2}{1+t^2} \right) = -t$$

ii) $\operatorname{cosec} \theta (\cos \theta - 1) = -1 \therefore -t = -1$

$$\therefore t = \tan \frac{\theta}{2} = 1 \therefore \frac{\theta}{2} = 45^\circ \therefore \theta = 90^\circ$$

c) $2x^3 + 4x - 3 = 0 \therefore 2\alpha^3 + 4\alpha - 3 = 0$

$$\therefore 2\alpha^3 = -4\alpha + 3 \text{ and similar for } \beta^3 \text{ and } \gamma^3$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = (-4\alpha + 3 - 4\beta + 3 - 4\gamma + 3)/2$$

Q14) (Continued)

d) $\frac{x}{x-1} \geq 2, \therefore 2(x-1) \leq x, \therefore x-2 \leq 0,$

$$\therefore x \leq 2 \text{ \& } x \neq 1 \therefore 1 < x \leq 2$$

Q15)

a) $2 \times 4^x - 3 \times 2^x + 1 = 0$

$$\therefore 2 \times (2^2)^x - 3 \times 2^x + 1 = 0$$

$$\therefore 2 \times (2^x)^2 - 3 \times 2^x + 1 = 0, \text{ Let } u = 2^x$$

$$\therefore 2u^2 - 2u + 1 = 0, \therefore (2u-1)(u-1) = 0$$

$$\therefore u = 2^x = \frac{1}{2} \text{ or } 1, \therefore x = -1 \text{ or } 0$$

b) TRIANGLE

i) all the vowels next to each other:

$$2! \times 3! \times 5! = 1440$$

ii) consonants are in the two end positions:

Choose the first and last letters from the 5

consonants, then arrange the remaining 6 letters

$$\text{between them in } (5 \times 4) \times 6! = 14400$$

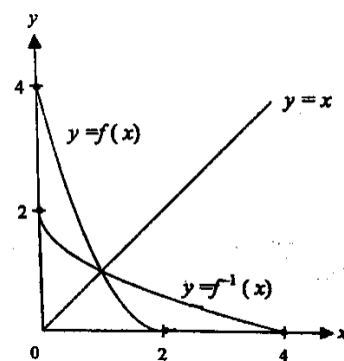
c) $f(x) = (x-2)^2, 0 \leq x \leq 2$

i) $y = (x-2)^2, 0 \leq x \leq 2 \text{ and } 0 \leq y \leq 4$

$$\therefore \sqrt{y} = |x-2| \therefore \sqrt{y} = 2-x \therefore x = 2 - \sqrt{y}$$

$$\therefore f^{-1}(x) = 2 - \sqrt{x}, \therefore \text{Domain } 0 \leq x \leq 4 [0,4]$$

ii)



d) $\sec x + \tan x = k - \textcircled{1}$

$$1 + \tan^2 x = \sec^2 x$$

$$\therefore \sec^2 x - \tan^2 x = 1$$

$$\therefore (\sec x + \tan x)(\sec x - \tan x) = 1$$

$$\therefore k \times (\sec x - \tan x) = 1$$

$$\therefore \sec x - \tan x = \frac{1}{k} - \textcircled{2}$$

$$\therefore \textcircled{1} + \textcircled{2} = 2 \sec x = k + \frac{1}{k}$$

$$\therefore \sec x = \frac{1}{2} \left(k + \frac{1}{k} \right)$$